

1 Safety

Read this manual before the unit is started for the first time. The unit delivers compressed air at a pressure that can injure; isolate it from the air system and lock the supply off before any part of it is opened.

Only a trained technician may change a controller setting or open the oil circuit. The oil and the air end stay hot for a long time after a stop.

3 Controller messages

The controller shows one message at a time. A warning leaves the unit running, a shutdown warning announces a limit that is close, and a shutdown stops the unit and inhibits a restart until it is reset by hand.

Code	Type	Message	Trigger	Threshold	Delay	Reset
W101	Warning	Line pressure low	Low-pressure switch closed in any state	7.0 bar (102 psi)	10 s	Automatic
W102	Warning	Continuous load time exceeded	Loaded without reaching cut-out	10 min	2 min	Automatic
W104	Warning	Oil temperature high	Oil temperature above the warning setting while running	75 °C (167 °F)	5 min	Automatic
W108	Warning	Motor starts per hour exceeded	More starts in one hour than the setting allows	6 per hour	0 s	Automatic
X201	Shutdown warning	Oil temperature very high	Oil temperature above the shutdown warning setting while running	85 °C (185 °F)	30 s	Automatic
S301	Shutdown	Oil temperature shutdown	Oil temperature above the shutdown setting while running	95 °C (203 °F)	5 s	Manual

4 Programmable settings

The controller refuses a value outside the range below, and it refuses a value that would put two settings in the wrong order.

Parameter	Unit	Min	Default	Max
Cut-in pressure	bar	5.0	8.0	9.5
Cut-out pressure	bar	6.0	10.0	11.0
Oil temperature warning	°C	60	75	90
Maximum starts per hour	per hour	1	6	12

8 Problem solving

8.1 Before you start

Work from what the unit shows, not from what it is said to do. Read the controller message list first, then the signals of the pneumatic panel, and only then open a component. A message names a limit that was crossed; it does not name the cause, and the same message follows from several of the causes listed in 8.2.

Let the unit run through one full load and unload cycle before a reading is taken. Oil temperature settles slowly, so a reading taken within ten

minutes of a start says little about the cooling circuit.¹

- Note the controller message, the state of the unit and the hour reading before anything is changed.
- Take every pressure from the pneumatic panel, never from a temporary gauge in the distribution line.
- Change one thing at a time and repeat the reading after one full cycle.

8.2 Fault tables

Each table below lists the causes of one condition, the signals that move, what to check and what to do. A cause that explains more than one condition keeps the same fault id under each of them.

1. Do not exceed 95 °C (203 °F): above that limit the controller stops the unit and inhibits the restart.

Fault id	Possible cause	Subsystem	Signals	Checks	Remedy
8.2.1 Line pressure below setpoint <code>low_line_pressure</code>					
<code>downstream_air_leak</code>	Downstream air leak. Air escapes from the distribution line after the unit, and the unit makes the loss up. Raises W101.	distribution	Line pressure (P2) falls faster than usual while unloaded. Load cycle rate is higher.	1. Close the isolation valve at the reservoir outlet. 2. Watch line pressure (P2) for ten minutes with the unit stopped. 3. Walk the distribution line and listen at every joint.	Repair the leaking joint or fitting and put the line back in service.
<code>intake_filter_clogged</code>	Intake filter clogged. The filter element is loaded with dust and restricts the air the element draws. Raises W101.	intake unloading	Line pressure (P2) is low while loaded. Motor current (I1) is low while loaded.	1. Read the filter restriction indicator. 2. Inspect the element against a lamp. 3. Compare motor current (I1) with the value of a clean element.	Fit a new intake filter element and reset the service counter.
<code>dryer_purge_leak</code>	Dryer purge leak. The purge valve does not seat, so regeneration air escapes without stopping. Raises W101 and W102.	dryer	Purge pressure (P4) is high while loaded. Line pressure (P2) falls while loaded.	1. Listen at the purge silencer with the unit loaded. 2. Read purge pressure (P4) over one full tower period. 3. Check the purge valve seat and its pilot line.	Overhaul or replace the purge valve and repeat the check.
8.2.2 Compressor starts and loads too often <code>frequent_cycling</code>					
<code>downstream_air_leak</code>	Downstream air leak. The same leak makes the unit load again soon after it unloads. Raises W108.	distribution	Load cycle rate is higher. Line pressure (P2) falls faster than usual while unloaded.	1. Record the number of starts in one hour. 2. Close the outlet isolation valve and watch line pressure (P2). 3. Walk the distribution line and listen at every joint.	Repair the leaking joint or fitting and put the line back in service.
<code>reservoir_isolation_valve_closed</code>	Reservoir isolation valve closed. Part of the reservoir volume is shut off, so the unit cycles on a small volume. Raises W108.	reservoirs	Load cycle rate is higher. Line pressure (P2) rises faster than usual while loaded.	1. Check the position of every reservoir isolation valve. 2. Compare the reservoir pressure with line pressure (P2). 3. Time one loaded run.	Open the isolation valve and lock it in the open position.
8.2.3 Oil temperature high <code>oil_temperature_high</code>					
<code>oil_cooler_fouled</code>	Oil cooler fouled. The cooler matrix is blocked with dust, so it rejects less heat than it	cooling	Oil temperature (T1) rises gradually in every state.	1. Inspect the cooler matrix against a lamp.	Clean the cooler matrix with the cleaning kit and repeat the reading.

Fault id	Possible cause	Subsystem	Signals	Checks	Remedy
	should. Raises W104 and X201.			2. Check the cooling air path for obstructions. 3. Read oil temperature (T1) after one hour at full load.	
high_ambient_temperature	No fault: high ambient temperature. The cooling air is warmer than the reference condition, so the oil settles higher. Raises W104.	cooling	Oil temperature (T1) is high in every state.	1. Read the temperature of the cooling air at the inlet. 2. Compare it with the reference conditions. 3. Check that warm exhaust air is not drawn back in.	Improve the ventilation of the machine room or shade the air inlet.
oil_filter_clogged	Oil filter clogged. The filter element restricts the oil flow to the airend, so less heat leaves with the oil. Raises W104 and S301.	oil	Oil temperature (T1) rises gradually while loaded.	1. Read the filter restriction indicator. 2. Check the hours since the last oil filter change. 3. Read oil temperature (T1) at full load.	Fit a new oil filter element and reset the service counter.

9 Technical data

The tags below are the signals the controller publishes. The source column names the column of the public dataset the replay reads.

Tag	Label	Description	Unit	Kind	Source column
discharge_pressure	P1	Pressure in the oil separator vessel at the airend discharge	bar	analog	TP2
line_pressure	P2	Line pressure at the pneumatic panel, the regulation signal	bar	analog	TP3
purge_pressure	P4	Pressure in the dryer purge line after the purge valve	bar	analog	DV_pressure
oil_temperature	T1	Compressor oil temperature at the airend outlet	°C	analog	Oil_temperature
motor_current	I1	Current of one motor phase	A	analog	Motor_current